

AHMED SHAHNEWAZ SIRAJ

CONTACT

☎ (513) 593-4779

✉ siraja@rpi.edu

📍 Troy, NY 12180

🌐 [linkedin.com/in/ahmed-shahnewaz-siraj](https://www.linkedin.com/in/ahmed-shahnewaz-siraj)

SKILLS

Power Electronics

- High-power DC-DC and DC-AC converter design (buck, boost, full-bridge, inverter topologies)
- IGBT and SiC device characterization, switching loss and DPT characterization
- Thermal stress analysis and physics-of-failure reliability assessment
- EMI-aware PCB layout, parasitic minimization, and gate-drive dv/dt control
- Kelvin force/sense measurement architecture for precision device characterization

Control & Modeling

- State-space modeling and frequency-domain control loop compensation
- Closed-loop current and voltage regulation with digital control implementation
- MATLAB/Simulink, PLECS, PSIM, and LT-spice simulation

Embedded Systems

- C/C++ firmware for power converter control
- TI C2000 DSP: ePWM, ADC-triggered ISR loops, SPI/SCI communication, telemetry
- Real-time embedded control architecture for automated stress testing systems

Hardware & Test Tools

- Multilayer PCB schematic capture and layout in Altium Designer
- Semiconductor static characterization using Keysight B1505A
- High-speed switching and spectrum analysis using oscilloscope-based measurements
- Inverter hardware integration, validation, and systematic debugging

PUBLICATIONS

- **A. Siraj** et al., “Reliability Database for Wide Bandgap Power Semiconductors and Module Packaging via Comprehensive Survey and Standardization,” 2025 IEEE IAS Annual Meeting, Taipei, Taiwan, 2025.
- **A. Siraj** et al., “Integrated AC Power Cycling Platform with Automated Characterization for T-Type Power Module in Photovoltaic Applications,” 2023 IEEE ECCE, Nashville, TN, 2023.
- **A. Siraj** et al., “Design and Implementation of Automated Characterization of T-type based Power Module for PV Inverter Reliability Assessment,” 2022 IEEE ECCE, Detroit, MI, USA, 2022.

SUMMARY

Ph.D. candidate in electrical engineering with expertise in power electronics, embedded control, and reliability engineering for high-power conversion systems. Led the design and deployment of a 60 kW, 800 V active power cycling platform enabling physics-of-failure-driven lifetime validation of PV inverter T-type power modules. Demonstrated full-stack system development across multilayer PCB hardware, TI C2000 DSP firmware, and MATLAB-based automated characterization infrastructure. Former GE Fellow intern with multiple IEEE publications.

EXPERIENCE

September 2023 – Present*

January 2021 – April 2023**

Graduate Research Assistant

*Rensselaer Polytechnic Institute, Troy, New York**

*Clemson University, Clemson, South Carolina***

Project: A Versatile Reliability Test Platform for T-type Power Modules for Photovoltaic Applications

- Architected and deployed a 60 kW AC power cycling platform for 3-phase T-type IGBT modules (4 devices/module): 4-layer Altium PCB design, 800 V power stage, Kelvin-connected sensing and conditioning, gate drive circuitry, and TI C2000 DSP firmware (interrupt-driven ADC, 3-level PWM, SCI telemetry, and fault protection)
- Developed automated Double Pulse Test (DPT) platform at 400 V bus; extracted turn-on/turn-off switching energies across multiple current and temperature setpoints, processing ~20 test conditions per phase, without manual intervention
- Built automated static characterization pipeline using Keysight B1505A for T-type IGBT modules; measured I_C-V_{CE} , I_C-V_{GE} , V_f , I_{CES} , and I_{GES} from -20 to 60°C , improving forward voltage measurement accuracy by 18% over manual baseline

May 2023 – August 2023

Fellow Intern

GE Research, Niskayuna, New York

Project: High-Voltage Partial-Discharge Detection & Insulation Reliability

- Designed and deployed a high-voltage (10 kV) partial discharge detection platform for pulsed power systems; developed MATLAB-based automation for oscilloscope data acquisition and stochastic signal analysis of partial discharge events
- Built an automated long-duration voltage cycling system for insulation reliability and degradation monitoring

August 2018 – December 2020

Graduate Assistant

Miami University, Oxford, Ohio

- Built short-circuit transient test platform for Si IGBT, SiC cascode JFET, and SiC MOSFETs (TO-247); analyzed conducted EMI impact across repetitive fault conditions

Sep 2014 – Jul 2018

Assistant Deputy Manager (Technical)

Summit Testing & Calibration Laboratory, Dhaka, Bangladesh

- Diagnosed and troubleshoot industrial VFDs, motor drives, crane systems, switchgear, and PLC-based automation at a power plant facility

EDUCATION

Expected 2026 **Ph.D., Electrical Engineering** *RPI, Troy, NY*

Thesis: Advancing Reliability of Test Platforms for Power Modules in Solar Inverter Applications

2021 **M.S., Electrical Engineering** *Miami University, Oxford, OH*

Thesis: Impact of Repetitive Short Circuit Transients on the Conducted EMI of SiC and Si-Based Power Devices

2013 **B.Sc., Electrical & Electronic Engineering** *AIUB, Bangladesh*